

Supplementary Document A:

ClinicalEarlyCancerDF Pipeline for the CancerSEEK Case Study

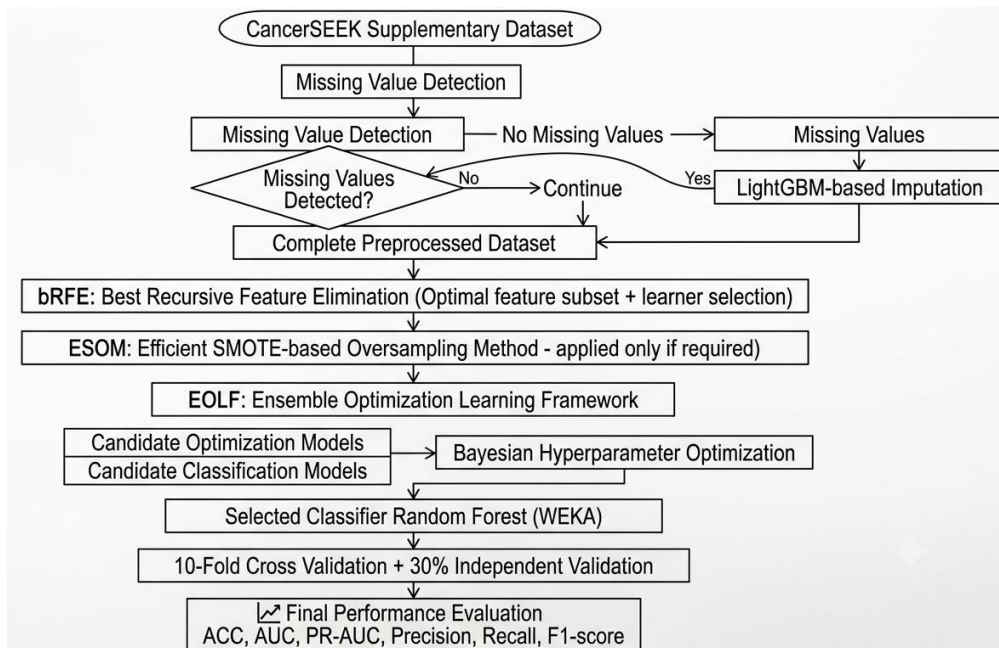


Figure 1. Selected Computational Pipeline for the CancerSEEK Cohort

ClinicalEarlyCancerDF is a generalized adaptive computational framework. The pipeline shown above represents the optimal workflow selected by the framework for the CancerSEEK cohort. Different datasets may result in different preprocessing strategies, feature subsets, optimization algorithms, and classification models depending on dataset characteristics.

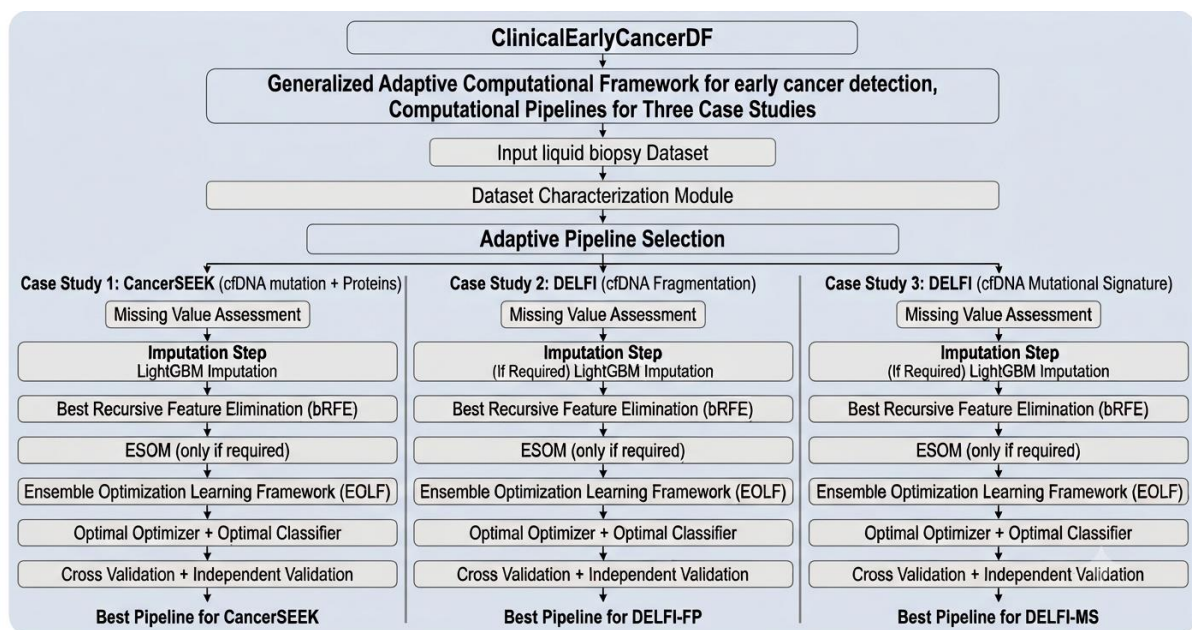


Figure 2. Generalized adaptive ClinicalEarlyCancer frameworks for three case studies

Purpose of the Case Studies

The purpose of the three case studies presented in Chapter 5 was not to demonstrate that a single machine learning algorithm consistently outperforms all previously published methods. Rather, they were designed to validate the proposed ClinicalEarlyCancerDF as a generalized adaptive computational framework capable of identifying an effective computational pipeline for heterogeneous liquid biopsy datasets. For each case study, the framework adaptively selected the preprocessing strategy, feature subset, optimization approach, and learning model through the integrated bRFE, ESOM (when required), and EOLF components according to the characteristics of the corresponding dataset. Consequently, the computational pipeline selected for the CancerSEEK cohort represents the optimal pipeline identified by the framework for that particular dataset, while different datasets may result in different pipeline configurations under the same generalized framework.

Supplementary Document B:

Table 1: Omega Score Ablation Study

Experiment	ACC	ROC_AUC	Interpretation
Omega Score only	62.76%	0.704	Omega score alone has limited discriminative ability.
Without Omega Score	96.92%	0.996	High performance is maintained using the remaining biomarker panel.

These results indicate that the performance of ClinicalEarlyCancerDF cannot be attributed solely to the published Omega score. Rather, the framework derives its predictive performance from the combined biomarker panel selected through the adaptive feature selection process.

Supplementary Document C:

Table 2: Selected Features for the CancerSEEK Cohort

Feature Category	Selected Features
cfDNA-derived	Omega Score, Mutant Fragments
Protein Biomarkers	CD44, CA-125, CEA, CA19-9, CYFRA21-1, GDF15, IL-6, IL-8, OPN, Prolactin, sEGFR, TGFa
Clinical Variables	Age, Sex

The final feature subset was identified using the Best Recursive Feature Elimination (bRFE) module of ClinicalEarlyCancerDF. The selected features comprise cfDNA-derived variables, protein biomarkers, and clinical variables that jointly maximize the predefined optimization objectives accuracy.

The objective of ClinicalEarlyCancerDF is different from that of the original clinical studies. The original publications primarily evaluated clinical assay performance under predefined protocols, whereas ClinicalEarlyCancerDF investigates whether an adaptive computational framework can identify an optimized analysis pipeline using the published biomarker measurements. Consequently, the reported performance reflects computational optimization of the available biomarker dataset rather than improvements to the underlying laboratory assay itself.

Additional verification of the adaptive framework performance:

To evaluate whether the observed discriminative performance depends on a particular classifier, an additional optimized learning pipeline (Bagging+A1DE) was evaluated with 10-fold cross-validation on the same processed CancerSEEK dataset. This independently achieved 99.17% accuracy and ROC AUC 0.9992, indicating that the high discrimination is not exclusive to a single learning algorithm but can also be achieved by alternative optimized pipelines identified within the proposed framework.